Lab Report: Identification and Management of Running Services

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System Environment: Kali Linux (Virtual Machine)

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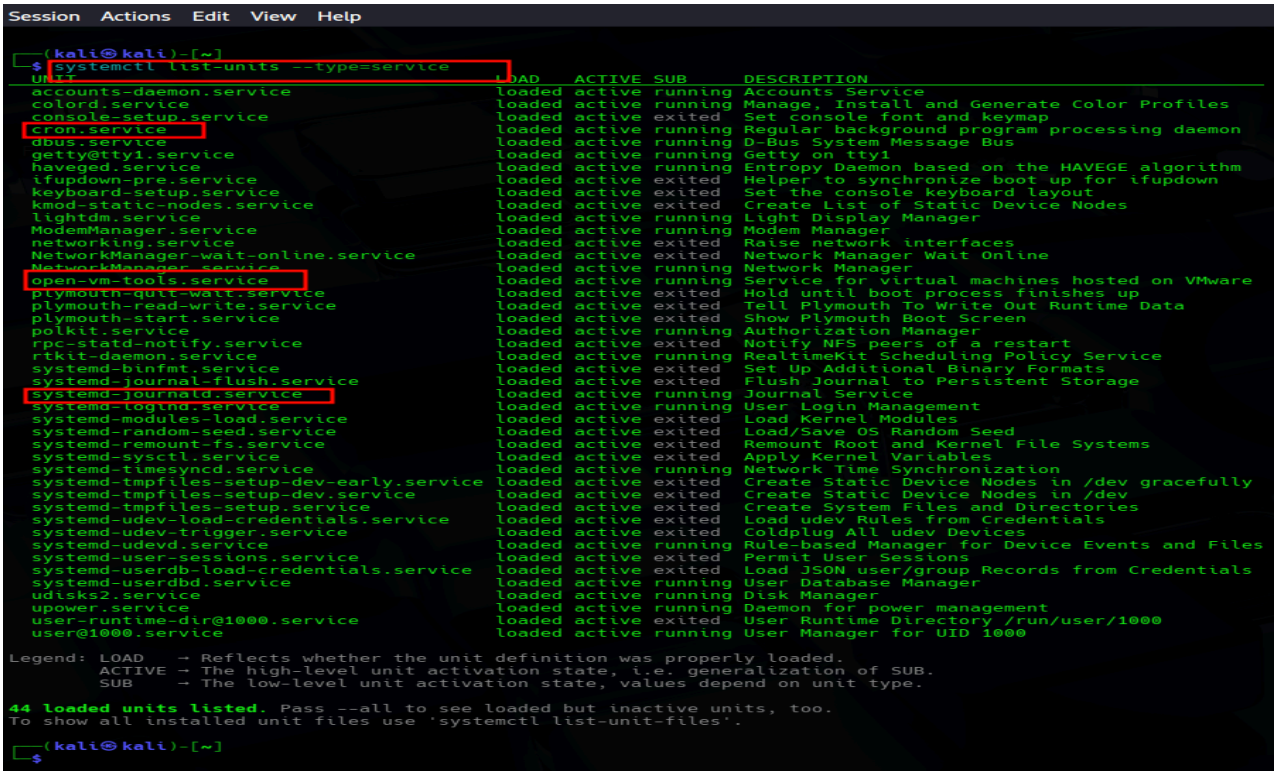
1. Objective

To identify active system services using command-line tools, select specific targets for analysis, and demonstrate the ability to safely stop and restart a service to verify process control.

2. Identified Active Services

Using the command `systemctl list-units --type=service`, the following three active services were identified from the system output:

- **`cron.service`**: The system job scheduler (currently active/running).
- **`systemd-journald.service`**: The system logging service (currently active/running).
- **`open-vm-tools.service`**: VMware utility service for host-guest integration (currently active/running).



```
Session   Actions   Edit   View   Help

┌──(kali㉿kali)-[~]
└─$ systemctl list-units --type=service
  UNIT                                      LOAD   ACTIVE SUB     DESCRIPTION
  accounts-daemon.service                   loaded active running Accounts Service
  colord.service                            loaded active running Manage, Install and Generate Color Profiles
  console-setup.service                     loaded active exited  Set console font and keymap
  cron.service                              loaded active running Regular background program processing daemon
  dbus.service                              loaded active running D-Bus System Message Bus
  getty@tty1.service                        loaded active running Getty on tty1
  haveged.service                           loaded active running Entropy Daemon based on the HAVEGE algorithm
  ifupdown-pre.service                      loaded active exited  Helper to synchronize boot up for ifupdown
  keyboard-setup.service                    loaded active exited  Set the console keyboard layout
  kmod-static-nodes.service                 loaded active exited  Create List of Static Device Nodes
  lightdm.service                           loaded active running Light Display Manager
  ModemManager.service                      loaded active running Modem Manager
  networking.service                        loaded active exited  Raise network interfaces
  NetworkManager-wait-online.service        loaded active exited  Network Manager Wait Online
  NetworkManager.service                    loaded active running Network Manager
  open-vm-tools.service                     loaded active running Service for virtual machines hosted on VMware
  plymouth-quit-wait.service                loaded active exited  Hold until boot process finishes up
  plymouth-read-write.service               loaded active exited  Tell Plymouth To Write Out Runtime Data
  plymouth-start.service                    loaded active exited  Show Plymouth Boot Screen
  polkit.service                            loaded active running Authorization Manager
  rpc-statd-notify.service                  loaded active exited  Notify NFS peers of a restart
  rtkit-daemon.service                      loaded active running RealtimeKit Scheduling Policy Service
  systemd-binfmt.service                    loaded active exited  Set Up Additional Binary Formats
  systemd-journal-flush.service             loaded active exited  Flush Journal to Persistent Storage
  systemd-journald.service                  loaded active running Journal Service
  systemd-logind.service                    loaded active running User Login Management
  systemd-modules-load.service              loaded active exited  Load Kernel Modules
  systemd-random-seed.service               loaded active exited  Load/Save OS Random Seed
  systemd-remount-fs.service                loaded active exited  Remount Root and Kernel File Systems
  systemd-sysctl.service                    loaded active exited  Apply Kernel Variables
  systemd-timesyncd.service                 loaded active running Network Time Synchronization
  systemd-tmpfiles-setup-dev-early.service  loaded active exited  Create Static Device Nodes in /dev gracefully
  systemd-tmpfiles-setup-dev.service        loaded active exited  Create Static Device Nodes in /dev
  systemd-tmpfiles-setup.service            loaded active exited  Create System Files and Directories
  systemd-udev-load-credentials.service     loaded active exited  Load udev Rules from Credentials
  systemd-udev-trigger.service              loaded active exited  Coldplug All udev Devices
  systemd-udevd.service                     loaded active running Rule-based Manager for Device Events and Files
  systemd-user-sessions.service             loaded active exited  Permit User Sessions
  systemd-userdb-load-credentials.service   loaded active exited  Load JSON user/group Records from Credentials
  systemd-userdbd.service                   loaded active running User Database Manager
  udisks2.service                           loaded active running Disk Manager
  upower.service                            loaded active running Daemon for power management
  user-runtime-dir@1000.service             loaded active exited  User Runtime Directory /run/user/1000
  user@1000.service                         loaded active running User Manager for UID 1000

Legend: LOAD   → Reflects whether the unit definition was properly loaded.
        ACTIVE → The high-level unit activation state, i.e. generalization of SUB.
        SUB    → The low-level unit activation state, values depend on unit type.

44 loaded units listed. Pass --all to see loaded but inactive units, too.
To show all installed unit files use 'systemctl list-unit-files'.

┌──(kali㉿kali)-[~]
└─$
```

3. Procedure Log & Observations

Step 1: Status Check

- **Action:** Check the initial status of the target service (cron).
- **Command:** `systemctl status cron`
- **Observation:** Service status confirmed as active (running).

```
┌──(kali㉿kali)-[~]
└─$ systemctl status cron
● cron.service - Regular background program processing daemon
     Loaded: loaded (/usr/lib/systemd/system/cron.service; enabled; preset: enabled)
     Active: active (running) since Thu 2026-02-12 05:02:36 EST; 2h 31min ago
 Invocation: 9ef1ac5c1ef3472a950dd5a7707693d7
       Docs: man:cron(8)
   Main PID: 1008 (cron)
      Tasks: 1 (limit: 4413)
     Memory: 584K (peak: 3.7M)
        CPU: 328ms
     CGroup: /system.slice/cron.service
             └─1008 /usr/sbin/cron -f

Feb 12 07:09:01 kali CRON[63605]: pam_unix(cron:session): session closed for user root
Feb 12 07:15:01 kali CRON[66580]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
Feb 12 07:15:01 kali CRON[66582]: (root) CMD (command -v debian-sa1 > /dev/null && debian-sa1 1 1)
Feb 12 07:15:01 kali CRON[66580]: pam_unix(cron:session): session closed for user root
Feb 12 07:17:01 kali CRON[67555]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
Feb 12 07:17:01 kali CRON[67557]: (root) CMD (cd / && run-parts --report /etc/cron.hourly)
Feb 12 07:17:01 kali CRON[67555]: pam_unix(cron:session): session closed for user root
Feb 12 07:25:01 kali CRON[71443]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
Feb 12 07:25:01 kali CRON[71445]: (root) CMD (command -v debian-sa1 > /dev/null && debian-sa1 1 1)
Feb 12 07:25:01 kali CRON[71443]: pam_unix(cron:session): session closed for user root

┌──(kali㉿kali)-[~]
└─$
```

Step 2: Service Termination

- **Action:** Stop the target service (cron).
- **Command:** `sudo systemctl stop cron`
- **Observation:** The service stopped immediately. A subsequent status check showed the state as inactive (dead).

```
┌──(kali㉿kali)-[~]
└─$ sudo systemctl stop cron
[sudo] password for kali:

┌──(kali㉿kali)-[~]
└─$
```

```
┌──(kali㉿kali)-[~]
└─$ systemctl status cron
○ cron.service - Regular background program processing daemon
     Loaded: loaded (/usr/lib/systemd/system/cron.service; enabled; preset: enabled)
     Active: inactive (dead) since Thu 2026-02-12 07:39:03 EST; 1min 15s ago
   Duration: 2h 36min 27.035s
 Invocation: 9ef1ac5c1ef3472a950dd5a7707693d7
       Docs: man:cron(8)
    Process: 1008 ExecStart=/usr/sbin/cron -f $EXTRA_OPTS (code=killed, signal=TERM)
   Main PID: 1008 (code=killed, signal=TERM)
   Mem peak: 3.7M
        CPU: 349ms

Feb 12 07:25:01 kali CRON[71443]: pam_unix(cron:session): session closed for user root
Feb 12 07:35:01 kali CRON[76308]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
Feb 12 07:35:01 kali CRON[76310]: (root) CMD (command -v debian-sa1 > /dev/null && debian-sa1 1 1)
Feb 12 07:35:01 kali CRON[76308]: pam_unix(cron:session): session closed for user root
Feb 12 07:39:01 kali CRON[78273]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
Feb 12 07:39:01 kali CRON[78275]: (root) CMD (   [ -x /usr/lib/php/sessionclean ] && if [ ! -d /run/systemd/system ]; then /usr/lib/php/sessionclean; fi)
Feb 12 07:39:01 kali CRON[78273]: pam_unix(cron:session): session closed for user root
Feb 12 07:39:03 kali systemd[1]: Stopping cron.service - Regular background program processing daemon...
Feb 12 07:39:03 kali systemd[1]: cron.service: Deactivated successfully.
Feb 12 07:39:03 kali systemd[1]: Stopped cron.service - Regular background program processing daemon.

┌──(kali㉿kali)-[~]
└─$
```

Step 3: Service Restoration

- **Action:** Restart the target service (cron).
- **Command:** `sudo systemctl start cron`
- **Observation:** The service successfully restarted. Status check confirmed state returned to active (running).

```
┌──(kali㉿kali)-[~]
└─$ sudo systemctl start cron

┌──(kali㉿kali)-[~]
└─$
```

```
┌──(kali㉿kali)-[~]
└─$ systemctl status cron
● cron.service - Regular background program processing daemon
     Loaded: loaded (/usr/lib/systemd/system/cron.service; enabled; preset: enabled)
     Active: active (running) since Thu 2026-02-12 07:41:59 EST; 1min 16s ago
 Invocation: 8312c0eb4d104ee9b4f40d5ce46c642c
       Docs: man:cron(8)
   Main PID: 79802 (cron)
      Tasks: 1 (limit: 4413)
     Memory: 348K (peak: 1.7M)
        CPU: 10ms
     CGroup: /system.slice/cron.service
             └─79802 /usr/sbin/cron -f

Feb 12 07:41:59 kali systemd[1]: Started cron.service - Regular background program processing daemon.
Feb 12 07:41:59 kali cron[79802]: (CRON) INFO (pidfile fd = 3)
Feb 12 07:41:59 kali cron[79802]: (CRON) INFO (Skipping @reboot jobs -- not system startup)

┌──(kali㉿kali)-[~]
└─$
```

4. Conclusion

The lab successfully demonstrated the ability to enumerate running services and control their state using the `systemctl` utility. The `cron` service was safely toggled off and on without error, verifying administrative privileges and proper service management syntax.